

SHAYAN IRANIPOUR

Curriculum Vitae

✉ siranipour.io

Experience

Isomorphic Labs — Senior Research Engineer

Feb 2025–present

- Lead development of the ADMET model for drug property prediction, owning the pretrain and finetune pipelines end-to-end.
- Scaled distributed training to large-scale multi-node deployments; built data parallelism and sequence sharding capabilities; established scaling laws across pretrain and finetune stages.
- Team expert in JAX, Flax, and XLA; experienced in writing distributed training code across large accelerator pools using manual mode sharding; extensive experience with transformer and GNN architectures; profiled and optimised model throughput using xprof; the most senior research engineer on the team.
- Worked at the intersection of research and systems engineering, driving significant gains in step time and internal evaluation metrics.

Xantium Group — Quantitative Researcher

Feb 2022–Dec 2024

- Built and deployed production ML models as trading signals over large, noisy datasets; worked across both the signal research stack and the core trading system, maintaining mission-critical Python code paths handling volumes in excess of \$1B.
- Developed an end-to-end ML research platform enabling the research team to rapidly prototype, iterate, and deploy models to production.

NNPDF Collaboration — PhD Researcher

Oct 2018–Mar 2022

- Applied deep learning to high-precision determinations of proton structure; authored 7 peer-reviewed papers with over 1,100 citations and presented at international conferences.
- Led migration from legacy C++ to TensorFlow/Keras, open-sourcing the project; experienced in distributed training and HPC resource management.

Education

University of Cambridge

- PhD — Department of Applied Mathematics and Theoretical Physics 2018–2022
Deep learning for particle physics. Smith-Knight & Rayleigh-Knight prize.
- MMath, Mathematics — Distinction 2017–2018
Part III of Mathematics Tripos. Specialised in theoretical physics.
- BA, Natural Sciences — Double 1st 2014–2017
Specialised in physics and mathematics. Sidney Sussex academic scholar.

Additional Skills

Technologies: Highly experienced with Python and scientific computing libraries. Good understanding of Docker, Kubernetes, GCP, Apache Beam, CI/CD, Slurm, SQL, and Redis. Familiar with C/C++, build systems, and Rust.

Libraries: Strong experience in JAX, Flax, PyTorch, TensorFlow, Keras, NumPy, Pandas/Polars, sklearn, XGBoost, Numba, matplotlib, xarray, dask, and SciPy.

Professional: Presented at international academic conferences and authored research in academic journals and internal reports. Experienced working across large codebases with teams across different timezones. Part of the hiring pipeline for developers and researchers.

Languages: Fluent in Farsi (Persian) and English (native speaker for both).